DGMS JSO GEOLOGY 2019 • GEOCHEMISTRY & CHEMICAL MINERALOGY

Question 98: Goldschmidt's Geochemical Classification: Chalcophile Silver

Thermodynamic Affinities, Electronic Configurations, and Element Partitioning

Q.98: Which one of the following is a chalcophile element?

(a) Oxygen

(b) Lithium

✓ **(c) Silver**

(d) Niobium

Correct Option: (c) Silver

1. Principles of Goldschmidt's Geochemical Classification

V. M. Goldschmidt (1922, 1937) classified the chemical elements according to their preferred geochemical affinities and distribution among the four primary phases formed during planetary differentiation:

- **Lithophile Elements ("Rock-loving" / Silicate-loving):** Elements with a strong chemical affinity for oxygen; they have outer electron configurations that readily form ionic bonds with O^{2-} and concentrate in silicate minerals and oxides in Earth's crust and mantle.
- **Chalcophile Elements ("Ore-loving" / Sulfur-loving):** Elements with intermediate electronegativities that form covalent bonds with polarizable S^{2-},

Se^{2-}, and Te^{2-} ions, concentrating predominantly in sulfide minerals and sulfide melt phases.

- **Siderophile Elements ("Iron-loving"):** Elements that form metallic bonds and concentrate in metallic iron-nickel phases (Earth's core).
- **Atmophile Elements ("Gas-loving"):** Highly volatile elements concentrated in the atmosphere and hydrosphere.

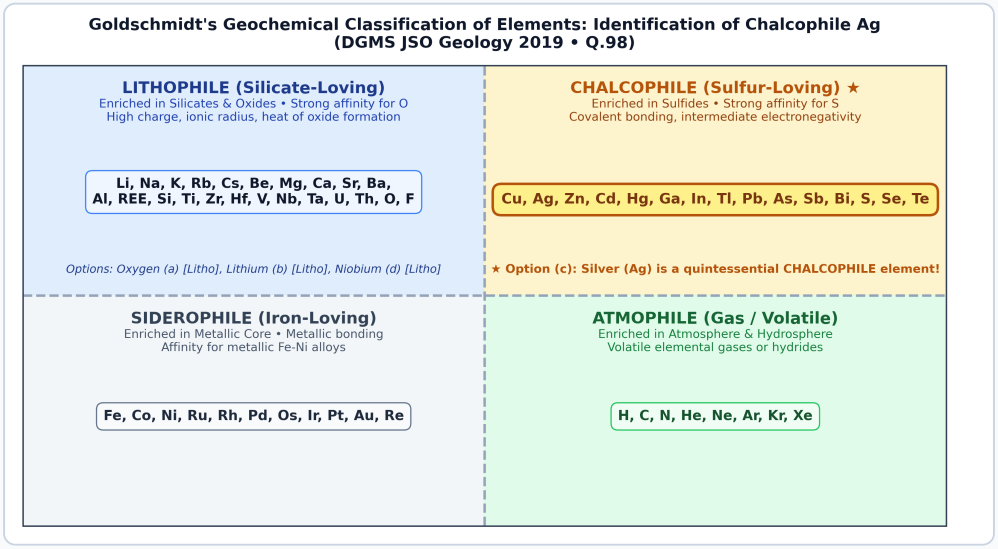


Figure 98.1: Summary of Goldschmidt's four geochemical groups (Lithophile, Chalcophile, Siderophile, Atmophile) highlighting Silver (Ag) as a classic chalcophile element in contrast to lithophile Oxygen, Lithium, and Niobium. Based on White, Geochemistry (Chapter 7).

2. Geochemical Characteristics of Silver (Ag)

Silver ([Kr] $4d^{10}$ $5s^{1}$, atomic number 47) is a quintessential **chalcophile element**:

- In igneous, metamorphic, and hydrothermal systems, Ag strongly partitions into sulfide melts and hydrothermal sulfide complexes with partition coefficients $D_{sulf/sil} > 100$.
- Silver forms primary economic sulfide and sulfosalt minerals such as **argentite/acanthite (Ag_2S)**, **pyrargyrite (Ag_3SbS_3)**, **proustite (Ag_3AsS_3)**, and tetrahedrite-freibergite ($(Ag,Cu)_{12}Sb_4S_{13}$).
- It also occurs in substantial solid solution in galena (PbS) via coupled substitution: $Ag^{+} + Bi^{3+} \leftrightarrow 2\ Pb^{2+}$.

3. Systematic Option Evaluation

Option	Element	Goldschmidt Classification	Geochemical Affinity & Natural Occurrence
(a)	Oxygen (O)	**Lithophile / Atmophile**	**Incorrect.** Oxygen is the defining ligand of lithophile elements, constituting ~46% of Earth's crust and forming the silicate tetrahedron $[SiO_4]^{4-}$.
(b)	Lithium (Li)	**Lithophile**	**Incorrect.** Group 1 alkali metal with high ionic potential; substitutes for Mg^{2+} in micas and pyroxenes (e.g., spodumene $LiAlSi_2O_6$, lepidolite).
(c)	**Silver (Ag)**	**Chalcophile**	**Correct.** Strong affinity for sulfur; concentrates in sulfide ore bodies, sulfosalts, and native silver associated with hydrothermal veins.
(d)	Niobium (Nb)	**Lithophile**	**Incorrect.** High Field Strength Element (HFSE, pentavalent Nb^{5+}) that concentrates in oxides (columbite-tantalite $[(Fe,Mn)(Nb,Ta)_2O_6]$) and carbonatites/ alkaline rocks.

Key Geochemical Takeaway

Goldschmidt's chalcophile group comprises **S, Se, Te, Cu, Ag, Zn, Cd, Hg, Ga, In, Tl, Pb, As, Sb, Bi**. In mining exploration and ore geology, these elements form the basis of base-metal and precious-metal sulfide mineralization.

4. Standard Textbook References

- **William M. White**, *Geochemistry* (2nd Edition, Wiley-Blackwell, 2020), Chapter 7: "Trace Elements", Section 7.2: Goldschmidt's Classification, pp. 270–285.
- **Brian Mason & Carleton B. Moore**, *Principles of Geochemistry* (4th Edition, John Wiley & Sons), Chapter 3: "Thermodynamics and Geochemical Classification of Elements", pp. 45–62.
- **Gunter Faure**, *Principles and Applications of Geochemistry* (2nd Edition, Prentice Hall), Chapter 6: "The Distribution of Trace Elements in Igneous Rocks", pp. 118–135.